

CREATIVE MEDIA · 17 MIN READ

Kinetic AI Video & Mastered Audio Generation Pipeline

Automated 9:16 Vertical Video Production, Remotion React, and Broadcast Audio

Executive Architecture Summary: Comprehensive engineering playbook for programmatic 9:16 vertical video synthesis using Remotion React, Kokoro / Edge-TTS audio pipelines, word-level kinetic subtitle synchronization, and -14 LUFS broadcast mastering.

1. Programmatic 9:16 Vertical Architecture

Short-form vertical video (TikTok, YouTube Shorts, Instagram Reels) demands 1080×1920 30FPS rendering with millisecond-accurate visual pacing, dynamic camera motion, and strict safe-zone adherence. The Remotion React engine enables programmatic rendering of React component trees into MP4 video files.

- Resolution & Frame Rate: 1080×1920 portrait canvas rendered at deterministic 30.0 FPS.
- Safe-Zone Enforcement: All interactive badges, pills, and kinetic text placed strictly between Y: 140px and Y: 1680px.
- Zero AI Slop Doctrine: Authentic UI captures, product screencasts, and clean glassmorphism preferred over generic stock bokeh.
- Direct Desktop Delivery: Rendered outputs exported directly to C:\Users\futur\Desktop\.

2. The Audio Pipeline & YouTube Audio Library Standard

High-retention video relies heavily on driving rhythmic music and punchy sound design. Audio beds and sound effects must be normalized to prevent voice-band frequency clash.



3. Kinetic Captions & Whisper Word-Level Timestamps

Transcribe voiceover tracks with OpenAI Whisper or Kokoro phoneme timings to generate precise per-word timing arrays. Render dynamic glowing highlight boxes with electric gold (#FACC15) on the currently spoken word.

```
// Remotion React Kinetic Captions Component
import { interpolate, useCurrentFrame } from 'remotion';

export const KineticWord = ({ word, startFrame, endFrame }) => {
  const frame = useCurrentFrame();
  const isActive = frame ≥ startFrame && frame ≤ endFrame;
  const scale = isActive ? interpolate(frame, [startFrame, startFrame + 3], [0.95, 1.08], {
    extrapolateRight: 'clamp' }) : 1.0;

  return (
    <span style={{
      transform: `scale(${scale})`,
      display: 'inline-block',
      color: isActive ? '#FACC15' : '#FFFFFF',
      fontWeight: isActive ? 900 : 700,
      textShadow: isActive ? '0 0 18px rgba(250, 204, 21, 0.65)' : '0 2px 8px rgba(0,0,0,0.8)',
      transition: 'color 0.1s ease'
    }}>
      {word}{',' '}
    </span>
  );
};
```

- MasterBroadcast Loudness: ffmpeg loudnorm pass enforcing integrated -14 LUFS (TP: -1.5, LRA: 8).
- Dynamic Ducking: Music bed automatically attenuated to 32% volume during voiceover segments.

4. Automated Render Farm & Desktop Pipeline

Execute headless rendering through Remotion CLI wrapped in non-interactive PowerShell scripts. Renders execute with GPU acceleration and parallel chunk encoding for sub-3-minute export speeds.

- GPU Chunk Encoding: NVENC / QuickSync acceleration handles 60-second video renders in under 90 seconds.
- Automated Thumbnail Extraction: Crisp 1080×1920 poster frame automatically generated at peak hook moment (T: 2.2s).